

STANDARD OPERATING PROCEDURE FOR 4 HI-MILL

4 HI-Mill OPERATION

- Relevant personal protective equipment to be used by operator.
- Go through rolling plan
- Check –up whether proper work rolls are available.
- Identify coil with coil /sticker and check width and gauge according to plan.
- Transfer the selected coil from awaiting rolling area with EOT crane to pay off reel skid.
- Load the coil from skid to pay – off reel with the help of coil car.
- Centre the coil and expand the pay-off mandrel
- Lower the hold-down (snubber) roll
- Cut the binding strap and feed the head end of the coil manually up to pinch roll by rotating pay-off reel in forward direction.
- Lower the pinch roll to hold the head end of the coil and lift the hold down (Snubber) roll
- Jog the pinch roll and bring the head end up to shear.
- In case the head is having uneven or bend edges shear the bad portion with the help of shear/gas cutting
- Ensure that roll gap between work –roll is more than the strip initial thickness
- Check the coil width, thickness at both edges of the width and enter the details in the 4 Hi- Mill Production Report.
- Lower the bridging apron (if applicable) and fed the strip up to entry side deflector roll
- Open the guide more than the strip width and set the top and bottom wooden drag-board
- Operator should arrange wood box in milling house before entry of the sheet and check the supply of the coolant in the housing mill
- By Jogging the “forward” main drive, feed the strip through drag -board and work roll gap with help of pinch roll or pay-off reel
- Tight the guide according to strip width with the help of wooden drag-board and down drag-board (Pneumatics pressure)
- Give the load on required to the first pass load on the end of strip and switch on the pay-off tension
- the strip mover rolls to delivery tension reel operate
- the mill in thread mode till the strip moves from roll bite to delivery tension reel
- Ensure levelling is proper by adjusting individual screw.
- Feed the strip to the delivery mandrel gripper and “Expand” the mandrel. Ensuring head-end is properly inserted inside the gripper.
- Rotate the delivery, mandrel jog in Jog mode and Main drive so that around one wrap of coil is wound on the mandrel.
- Stop the jogging and switch on delivery tension.
- Feed pass schedule in PLC Programme
- Select first pass from delivery side panel board
- Apply required load. Start the coolant spray and start the mill.
- Check thickness by measuring with Micro meter, compute the difference observed during physical checking with non-contact gauge measuring unit.
- The rolling should continue as per plan schedule till we get required thickness's
- Put a binding strap around the coil, give identification with coil card/ sticker and sent to next work center.
- Quality inspection with hand instrument should not carried out while machine running
- Write the production report.